

Simple Interface for Digital Sound Synthesis



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You can make a digital sound synthesizer, for experimentation with sound equipment or with peripheral devices, using special integrated circuits or digital-to-analogue converters (DACs). But here is a simple 8-input digital sound-synthesis circuit for producing audio from digital codes that can be easily interfaced with microcontroller or microprocessor boards having up to eight TTL/CMOS-level digital output pins.

Circuit and working

Fig. 1 shows circuit diagram of the simple interface for digital sound synthesis. It is built around low-power audio amplifier LM386 (IC1), 9V voltage regulator 7809 (IC2) and a bridge rectifier (BR1).

The 8-bit digital code (representing audio signal) input is applied at connector CON4. Resistors R1 through R8 and potmeter VR1 work as a simple DAC circuit. The relationships to get approximate values of resistors in the DAC are:

$R1 = 2R2 = 4R3 = 8R4 = 16R5 = 32R6 = 64R7 = 128R8$

Value of VR1 can be from 10% of R8 to 20% of R8.

The minimum value of each resistor (R1 through R8) depends on digital output levels. Capacitors C14 and C15, with switches S1 and S2 closed, filter the sound signals, and can be omitted, if not needed.

The audio input is amplified by LM386 low-power amplifier. The gain is set to 200. With 9V power supply,

Test Points

Test point	Details
TP0	0V
TP1	12V to 18V DC
TP2	9V DC

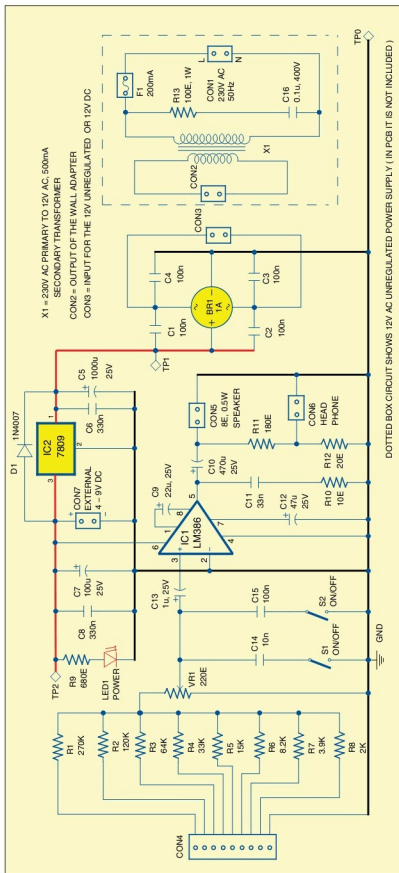


Fig. 1: Simple interface for digital sound synthesis